Yonatan Bason

Mechanical Engineer
Davie, FL, USA
+1 954-830-5509 | yonibason@gmail.com | linkedin.com/in/yonatan-bason-486ab7221

Summary

Mechanical engineer with strong foundations in structural mechanics, thermal systems, and mechanical design. Experienced in component and assembly-level development, hardware integration, and performance validation. Proven ability to translate specifications into manufacturable designs, perform structural and thermal evaluations, and support testing through iterative design refinement.

Highlights of Expertise

- Solid Mechanics & Structural Analysis
- Stress & Deformation Evaluation
- High-Pressure Systems (Up to 40 bar)
- Mechanical Assembly & Validation
- Failure Analysis
- CAD Modeling (SolidWorks, CREO)
- DFM & GD&T
- Finite Element Fundamentals
- Experimental Design & Testing
- Cross-Functional Collaboration

Professional Experience

Rosh MFM September 2026 -
Head of Research and Development (R&D) *Remote - New York, New York*

- Lead the R&D function for an early-stage biomedical device, owning technical direction from concept through bench-validated prototype — mechanical design (SolidWorks), design-for-manufacture, and iterative prototyping.
- Build first-principles physics and measurement models in Python (NumPy/Matplotlib) and scope nonlinear/hyperelastic FEA for soft-material and tissue-contact analysis to guide design decisions.
- Design and run bench validation (fixtures, protocols, calibration, pass/fail experiments) and define the BOM, R&D budget, sourcing strategy, and regulatory/biocompatibility framework (ISO 10993, sterilization, 510(k)/De Novo).

Arlo Industries (YC P26) March 2026 - September 2026
Engineer *San Fransisco, California*

- Lead R&D on complex hardware nodes, overseeing the end-to-end design, simulation, and integration of sensor-heavy platforms.
- Utilize CAD software for parametric modeling of high-stress mechanical assemblies and custom enclosures.
- Apply material science and 3D printing expertise to engineer functional, field-tested components using industrial-grade materials.
- Execute full-stack development, coordinating mechanical build requirements with software implementation to meet mission-critical tracking benchmarks.

Technion Rocketry Club Jun 2022 – Jan 2025
Propulsion Engineer *Haifa, Israel*

- Designed and modeled hybrid rocket propulsion assemblies in SolidWorks, developing components, subassemblies, and structural support hardware operating at chamber pressures up to 40 bar.
- Performed analytical structural and thermal evaluations (hoop stress, axial stress, thermal expansion) to verify design margins and requirements adherence.
- Generated detailed part and assembly drawings to support fabrication and integration.
- Participated in hardware builds, integration, and static fire testing, evaluating system performance and recommending iterative design modifications.
- Improved stiffness-to-weight ratio and structural robustness through design refinement based on test feedback.

Faculty of Mechanical Engineering, Technion Dec 2022 – Jan 2025
Undergraduate Researcher *Haifa, Israel*

- Performed full-field strain and deformation analysis using Digital Image Correlation (DIC) under dynamic loading conditions.
- Correlated experimental strain data with solid mechanics models to validate structural behavior and quantify load redistribution.
- Identified a 14% increase in horizontal force redirection through vector-based deformation analysis, improving understanding of load path behavior.
- Designed and documented custom mechanical fixtures to improve boundary condition repeatability and structural test consistency.
- Documented experimental setup, analytical methodology, and performance results to support reproducibility and future testing phases.

Alexion Pharmaceuticals (Rare Disease Unit of AstraZeneca; acquired LogicBio in 2022) 2021 – 2023
Engineering Intern (3 Terms) *Lexington, MA*

- Designed and prototyped mechanical hardware for a cost-efficient purification system, reducing process costs by 3× through structured design iteration and validation testing.

- Supported engineering scale-up of bioprocess systems (2L–50L), evaluating structural integrity, thermal performance, and fluid handling behavior under increased operational demands.
- Participated in equipment installation, performance verification, and troubleshooting, recommending mechanical design modifications to improve reliability and repeatability.
- Collaborated with cross-functional teams to generate documentation including procedures, test records, and updated design specifications.
- Improved assay throughput by 20% through thermal workflow optimization and systematic bottleneck analysis.

Education

Technion Israel Institute of Technology Haifa, Israel

B.Sc. Mechanical Engineering (Mechatronics, Robotics) *2020 – 2025*

- **Relevant Coursework:** Machine Design, Solid Mechanics, Dynamics, Heat Transfer, Thermodynamics, Fluid Mechanics, CAD

Technical Skills

- **Software:** MATLAB, Python, SolidWorks, CREO
- **Manufacturing:** CNC Machining, 3D Printing, Injection Molding
- **Languages:** English, Hebrew